

Dimensionality Reduction

Introduction

- ❑ In many learning problems, the datasets have large number of variables.
- ❑ Sometimes, the number of variables is more than the number of observations.
- ❑ For example, in many scientific fields such as
 - image processing,
 - time series analysis,
 - Internet search engines, and
 - automatic text analysis.

Introduction

- Statistical and machine learning methods have some difficulty when dealing with such high-dimensional data.
- Normally the number of input variables is reduced before the machine learning algorithms can be successfully applied.
- In statistical and machine learning, dimensionality reduction or dimension reduction is the process of reducing the number of variables under consideration by obtaining a smaller set of principal variables.

Dimensionality Reduction - Types

Dimensionality reduction may be implemented in two ways.

- Feature selection
- Feature extraction

Dimensionality Reduction - Types

Dimensionality reduction may be implemented in two ways.

- Feature selection
- Feature extraction

Feature Selection

In feature selection, we are interested in selecting **k features** out of the total **n features** that provide the most useful information, and we discard the remaining **(n – k) features**.

Dimensionality Reduction - Types

Feature Extraction

- In feature extraction, we aim to create a **new set of k features** that are formed by combining the original **n features**.
- These techniques can be **supervised or unsupervised**, depending on whether they use output (label) information or not.
- The most well-known and widely used **feature extraction methods** are **Principal Components Analysis (PCA)** and **Linear Discriminant Analysis (LDA)**.

Dimensionality Reduction - Measures of error

In both methods we require a measure of the error in the model. In regression problems, we may use the

- **Mean Squared Error (MSE)** or the
- **Root Mean Squared Error (RMSE)**

Dimensionality Reduction - Measures of error

Mean Squared Error (MSE)

MSE is the sum, over all data points, of the square of the difference between the predicted and actual target variables, divided by the total number of data points.

- If $y_1, y_2, \dots, y_n$ are the observed values and $\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n$ are the predicted values, then

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Misclassification Rate

In classification problems, the misclassification rate can be used as a measure of error.

It is defined as

$$\text{misclassification rate} = \frac{\text{number of misclassified examples}}{\text{total number of examples}}$$